

# TEJAS MELKOTE

AI/ML Engineer | Full-Stack Software Engineer

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## PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science (AI) engineer who ships production software and peer-reviewed research. Delivered 15+ pull requests to a live Go, Ruby on Rails, and Vue.js codebase at SuperAGI, and currently builds an AI construction-intelligence platform at TatvaOps serving real users on React and TypeScript. First author of an accepted Sci-FM poster at COLM 2026 on LLM calibration. Seeking AI/ML Engineer or Software Engineer roles building production systems.

## PROFESSIONAL EXPERIENCE

### Full-Stack / AI Engineer, TatvaOps

May 2026 - Present

Bengaluru, India

- Build and ship Vantage, an AI-powered construction knowledge platform (InSights, Radar, Forums, Studio, Eva AI) on React, TypeScript, and Vercel, serving live users.
- Shipped a verified vendor profile system with search, ratings, and pricing insights for contractor discovery.
- Built CMS-driven SEO content pipelines and analytics dashboards that grow high-intent organic traffic.

### Software Development Engineer Intern, SuperAGI

Jan 2026 - May 2026

Bangalore, India

- Delivered 15+ production pull requests into a full-stack Go, Ruby on Rails, and Vue.js codebase through a Jenkins and ArgoCD CI/CD pipeline.
- Built Email and WhatsApp marketing campaign features end to end, including campaign validation logic and a WhatsApp test-send capability used by marketing teams.
- Engineered background jobs and event streaming with Sidekiq, Redis, and Kafka, and executed database migrations safely against production data.
- Integrated CRM workflows and controller endpoints, owning features from ticket through deployment.

### AI / ML Intern, CirrusLabs Pvt Ltd

May 2025 - Jul 2025

Bengaluru, India

- Built LLM chatbots, AI agents, and ML pipelines that added AI capability to internal project tooling.
- Packaged and served models behind FastAPI and Streamlit interfaces so non-technical teams could use them directly.

## RESEARCH

### Algorithmic Blindness in Large Language Models: A Calibration Study

Accepted Poster

Sci-FM Workshop, COLM 2026 (non-archival)

- Evaluated 8 frontier models across 5,200+ algorithm runs and measured 15.9% calibrated coverage, evidencing systematic failure of LLM self-performance prediction.

## PROJECTS

- TatvaOps Vantage** | [Live](#) AI construction knowledge platform with CMS, forums, analytics dashboards, and AI-assisted content; React and TypeScript on Vercel.
- TastelQ, Construction Intelligence Network** | [Live](#) Swipe-first taste engine that learns a user's Design DNA from real spaces, builds a Taste Graph, and matches homeowners to vendors by aesthetic fit.
- INUIT, Conversational Luxury Footwear** | [Live](#) Storefront driven by an AI concierge that handles fit, styling, and ordering in conversation, replacing traditional checkout forms.
- AI Trip Planner** | [GitHub](#) Agentic planner orchestrating 7 external APIs (Weather, Tavily, Google Places, Groq) through a LangGraph workflow, with a FastAPI backend and Dockerized deployment.
- Machine Learning Portfolio** | [GitHub](#) Crop Yield Prediction (hybrid CNN-RNN-LSTM), Car Price Prediction (scikit-learn pipeline), Customer Churn and Segmentation (SVM, KNN, DBSCAN), and Whisper-based real-time Speech-to-Text.

## TECHNICAL SKILLS

**Programming Languages:** Python, Go, Java, C, C++, Ruby, JavaScript, TypeScript, SQL, HTML, CSS

**AI and Machine Learning:** Large Language Models, RAG, AI Agents, LangGraph, LangChain, PyTorch, TensorFlow, scikit-learn, NLP, Computer Vision, Model Calibration, Vector Databases

**Backend and Web:** FastAPI, Ruby on Rails, Go services, React, Vue.js, Node.js, Tailwind CSS, Streamlit, REST APIs

**Data and Infrastructure:** PostgreSQL, Redis, Kafka, Sidekiq, Docker, Jenkins, ArgoCD, Vercel, Git, GitHub, Power BI, Web Scraping

## EDUCATION

### Bachelor of Technology, Computer Science Engineering (Artificial Intelligence)

2022 - 2026

Manipal Institute of Technology, Bengaluru | CGPA: 8.38 / 10

Class XII, Narayana PU College, Bengaluru (90.33%), 2022 | Class X, Frank Anthony Public School, Bangalore (95%), 2020

## CERTIFICATIONS AND LEADERSHIP

Certifications: Career Essentials in Generative AI (Microsoft and LinkedIn); Software Engineering Job Simulation (J.P. Morgan, Forage); Junior Cybersecurity Analyst (Cisco); Algorithmic Toolbox and Oracle SQL Databases (Coursera).

Leadership: Class Representative (AI-C); PR Head, E-Cell; Operations Head, College Fest.

Languages: English (Native), Hindi (Proficient), Kannada (Advanced).